Agentic SABRE

An Uncertainty-Aware Multi-Agent Framework
for Adaptive Ransomware Detection

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Abstract

Ransomware detection demands evidence from multiple modalities — static PE structure, runtime hypervisor telemetry, and threat-intelligence discourse — yet no single detector dominates across all sample types. SABRE addresses this through a three-agent architecture: a **Semantic CNN** (RDset, 138,047 PE samples), a **Behavioural CNN** (RanSMAP, 751,588 hypervisor telemetry windows), and a **Knowledge CNN** (RansKnow, 200 MITRE ATT&CK samples). Each agent produces a probability and an uncertainty estimate via Monte Carlo Dropout. A **Decision MLP** fuses the six-dimensional score vector $[\hat{p}_{\text{sem}}, \hat{p}_{\text{beh}}, \hat{p}_{\text{know}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}}]$ (augmented with CTGAN and calibrated with temperature scaling) into a fused risk $\hat{p}$. An **uncertainty-aware triage policy**, tuned by Bayesian optimisation (TPE), maps $(\hat{p}, \sigma_{\max})$ to auto-contain / escalate / allow. Two new mechanisms extend the baseline: **per-agent peer consultation** (κ_{local}) adjusts uncertain agent scores before fusion, and an **expert feedback loop** enables analyst corrections to drive incremental MLP retraining. The fused system achieves **97.2% balanced accuracy, 100% TPR, 94.7% precision, and zero false negatives** on a 4,000-sample evaluation.

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1 Introduction

Ransomware remains one of the most operationally damaging classes of malware. Its continued evolution through polymorphism, living-off-the-land techniques, and behavioural obfuscation means that signature-based defences are insufficient. Modern detectors must adapt to out-of-distribution samples, handle evidence of varying quality, and support human-in-the-loop oversight.

Three challenges motivate the SABRE design:

1. **Heterogeneous evidence.** Static PE features, runtime storage and memory telemetry, and threat-intelligence discourse encode distinct aspects of ransomware behaviour. No single modality dominates across all sample types.
2. **Missing uncertainty signals.** Standard ML classifiers produce point predictions. An aggressive threshold creates false positives and operational disruption; a conservative one misses threats. Out-of-distribution samples make confidence values unreliable without explicit uncertainty quantification.
3. **Operational tractability.** A deployed triage system must support automated containment of high-confidence threats, analyst escalation for ambiguous cases, and a mechanism for analyst corrections to improve the system over time.

SABRE answers all three through a three-agent fusion pipeline with uncertainty-aware decision making, per-agent peer consultation, and an expert feedback loop.

2 Datasets

2.1 RDset — Semantic Agent

RDset contains **138,047 PE (Portable Executable) samples** (96,724 ransomware, 41,323 benign) with 57 structured static attributes per sample. Attributes covering import tables, section characteristics, header metadata, and byte-level statistics are encoded into text strings and embedded with a 384-dimensional BERT-style encoder, producing the input feature vectors for the Semantic CNN. Training uses a SMOTE-balanced split.

2.2 RanSMAP / RanSAP — Behavioural Agent

The behavioural dataset contains **751,588 hypervisor-level telemetry windows** (750,329 ransomware, 1,259 benign — a 596:1 class imbalance). Each window comprises 16 sliding-window statistics derived from storage I/O and memory access traces:

- Storage: average read/write throughput, read/write LBA variance, write entropy
- Memory: read/write/execute GPA variance; read/write/readwrite entropy; access counts

Features are normalised with a `StandardScaler` (critical for CNN performance; without it balanced-eval AUC degrades from 0.996 to 0.621). For training, the majority class is undersampled to 10,000 before SMOTE is applied to balance the classes.

2.3 RansKnow — Knowledge Agent

RansKnow comprises **200 labelled samples** derived from YouTube ransomware practitioner discourse. Each sample is represented as a 23-dimensional feature vector covering MITRE ATT&CK tactic mentions (Initial Access, Execution, Persistence, Privilege Escalation, Credential Access, Lateral Movement, Discovery, C&C, Exfiltration, Impact), tool mentions (Cobalt Strike, Mimikatz, PsExec, Rclone, MegaNZ, AnyDesk, TeamViewer, BloodHound), platform signals (Windows, Linux, ESXi), and aggregated tactic/tool counts.

3 System Architecture

Figure 1 shows the full pipeline. The system is organised into five conceptual stages: base detection, score pooling, decision fusion, triage policy, and the expert feedback loop.

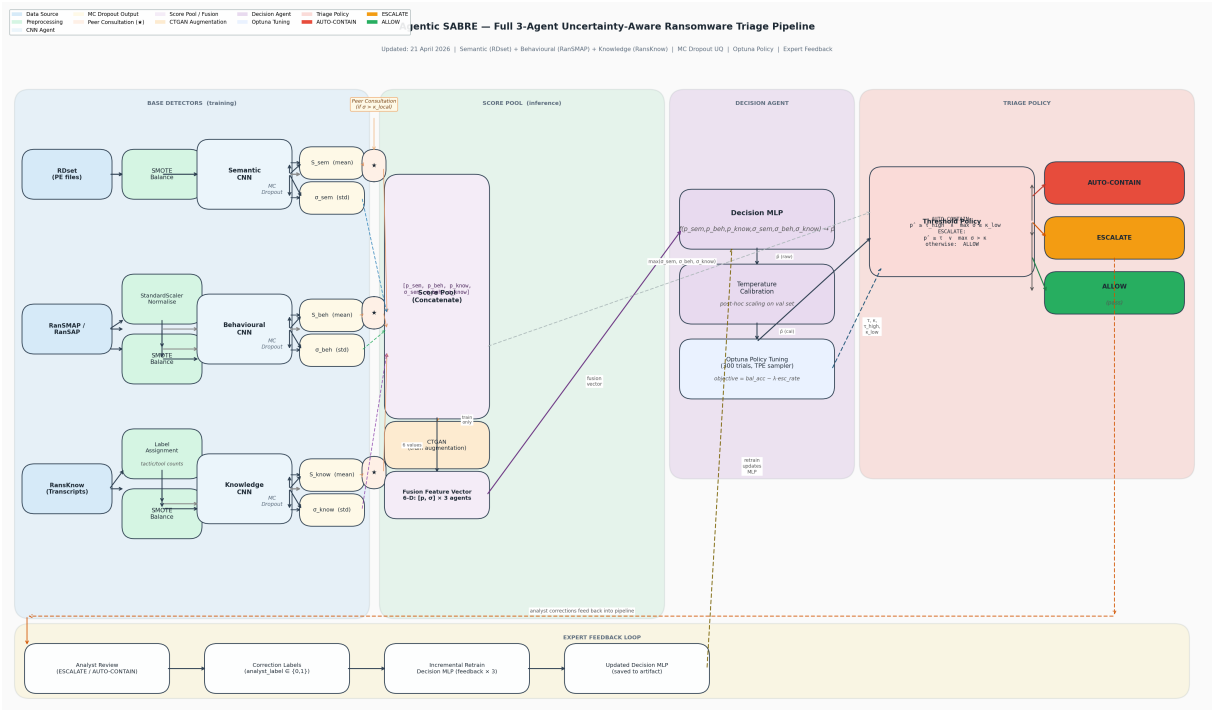


Figure 1: Full Agentic SABRE pipeline. Peer consultation gates (★) fire per agent when $\sigma_{\text{agent}} > \kappa_{\text{local}}$, adjusting the probability before it enters the Score Pool. The Expert Feedback Loop (bottom) routes analyst corrections back into incremental MLP retraining.

3.1 Base Detectors

All three agents share a common structure:

1. A CNN classifier trained on the agent's dataset (with SMOTE balancing and, for the behavioural agent, StandardScaler normalisation).
2. **Monte Carlo Dropout** uncertainty quantification at inference: the model is run T times with dropout active; the mean output is the probability $\hat{p}$ and the standard deviation is the uncertainty σ. The number of passes T is adaptive — sampling continues until σ converges (maximum $T = 50$).

3.2 Score Pool and Decision MLP

The six outputs $[\hat{p}_{\text{sem}}, \hat{p}_{\text{beh}}, \hat{p}_{\text{know}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}}]$ are concatenated into a **Fusion Feature Vector**. During training, CTGAN generates synthetic score-space samples to augment the training distribution. The **Decision MLP** $f : \mathbb{R}^6 \to [0, 1]$ maps this vector to a fused risk $\hat{p}$. Post-training **temperature scaling** calibrates $\hat{p}$ against a held-out validation set.

3.3 Triage Policy

The calibrated $\hat{p}$ and the maximum agent uncertainty $\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}})$ are passed to the triage policy:

Policy Rules

$$\begin{aligned} \text{auto-contain} &: \hat{p} \geq \tau_{\text{high}} \ \wedge \ \sigma_{\max} \leq \kappa_{\text{low}} \\ \text{escalate} &: \hat{p} \geq \tau \ \vee \ \sigma_{\max} > \kappa \\ \text{allow} &: \text{otherwise} \end{aligned}$$

The four thresholds $(\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}})$ are tuned by **Bayesian optimisation using the Tree-structured Parzen Estimator (TPE)** via Optuna (300 trials, objective: balanced accuracy $-\lambda\cdot$ escalation rate). Using $\sigma_{\max}$ ensures that any single uncertain agent can veto autonomous containment, preserving auditability.

4 Per-Agent Peer Consultation

Each agent has an independent **local uncertainty threshold** κ_{local}. At inference, after MC Dropout produces $(\hat{p}_i, \sigma_i)$, the orchestrator checks: if $\sigma_i > \kappa_{\text{local}}$, the agent's probability is adjusted using the inverse-uncertainty-weighted scores of itself and both peer agents:

$$w_j = \frac{1}{\sigma_j + \varepsilon}, \qquad \hat{p}_i^{\text{adj}} = \frac{\sum_{j \in \{i, \text{peers}\}} w_j \hat{p}_j}{\sum_j w_j} \tag{1}$$

The adjusted $\hat{p}_i^{\text{adj}}$ (not the original $\hat{p}_i$) enters the Score Pool. Critically, σ_i is **never modified** — the uncertainty veto operated by the global triage policy remains intact. The result records the consultation flag, raw probability, adjusted probability, and per-participant weights.

The default $\kappa_{\text{local}} = 1.0$ means consultation never fires unless explicitly configured, making the mechanism fully backwards-compatible.

5 Expert Feedback Loop

escalate and auto-contain results carry a UUID `sample_id`. Analysts submit corrections through the dashboard *Feedback* tab. Corrections are stored in an append-only JSON log containing: timestamp, agent scores, uncertainties, policy decision, expert label $\in \{0, 1\}$, and free-text reasoning.

When $N \geq 20$ corrections accumulate, *Trigger Retrain* becomes available. Retraining reconstructs the training distribution from the Decision MLP scaler statistics, synthesises a base set, then appends feedback rows with 3× sample weight. The MLP is retrained and the updated checkpoint saved. The operator may promote it manually or via `--promote`; TPE then re-tunes κ_{local} for each agent against the corrected labels.

6 Evaluation

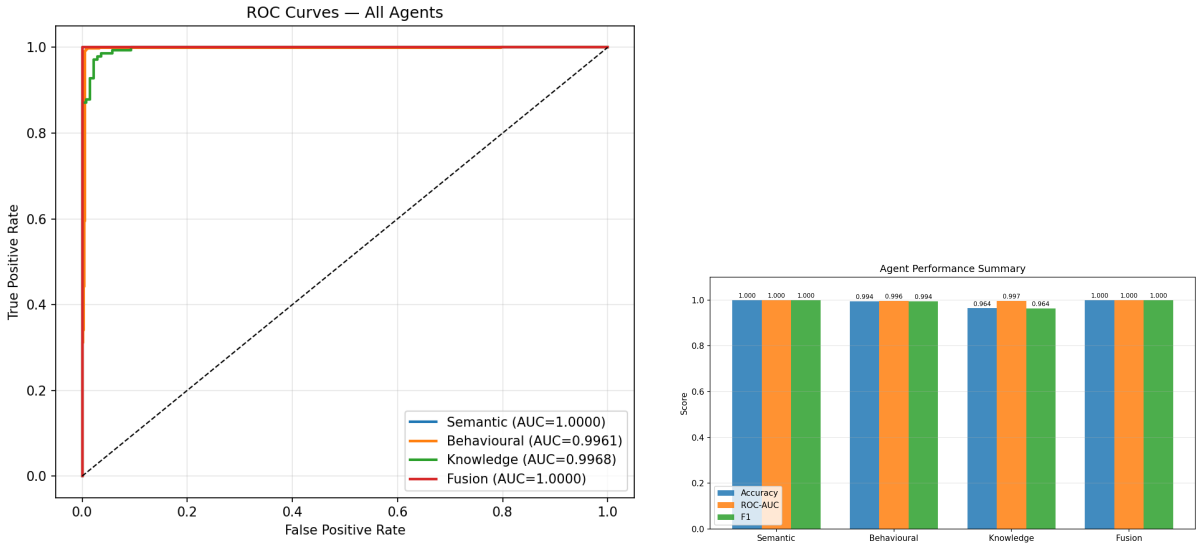
6.1 Per-Agent Results

Table 1 reports per-agent performance. Two evaluation regimes are used: *natural distribution* (same class ratio as the source dataset) and *balanced* (1,000 samples per class — equal-split evaluation, the harder test for discrimination ability).

Table 1: Per-agent and fused-system evaluation results.

Agent	Train N	Nat. Acc	Nat. AUC	Bal. Acc	Bal. AUC
Semantic	138,047	100.0%	1.000	100.0%	1.000
Behavioural†	751,588	99.6%	0.999	99.6%	0.996
Knowledge	200	97.4%	0.997	96.5%	0.995
Fused System	All	97.2% balanced acc		AUC 1.000	

† 596:1 class imbalance; SMOTE + StandardScaler applied during training.
Without StandardScaler, behavioural balanced AUC degrades to 0.621.



(a) ROC curves — all agents and fused system. **(b)** Accuracy, AUC, and F1 summary.

Figure 2: Agent and fusion evaluation metrics.

6.2 Fused System and Triage Policy

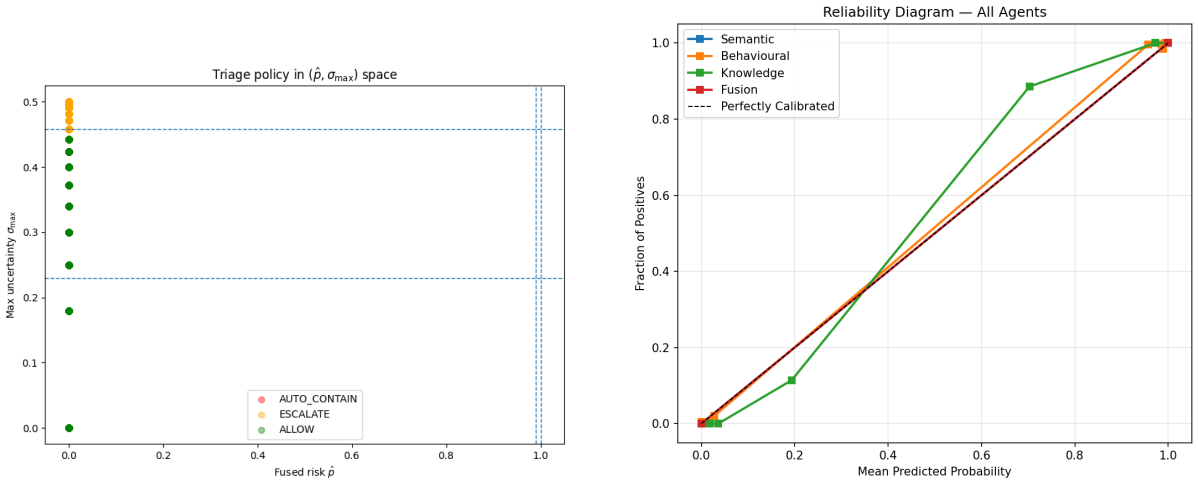
Table 2 compares the two tuned policy configurations on 4,000 balanced samples (2,000 ransomware, 2,000 benign).

Table 2: Triage policy performance — 4,000-sample balanced evaluation.

Policy	τ	κ	τ_h	κ_l	Bal. Acc	TPR	FPR	Esc. Rate
Default (auto)	0.396	0.196	0.905	0.195	97.2%	100%	5.6%	52.8%
Low escalation	0.233	0.197	0.854	0.197	97.6%	100%	4.8%	52.4%

Both policies: FN = 0. Esc. rate = fraction of samples routed to escalate.

Both policies achieve **zero false negatives** — no ransomware sample is allowed through. The 112 false positives (default policy) represent benign samples routed to escalate or auto-contain rather than allow; given the escalation rate, the majority of FPs are routed to analyst review rather than autonomous isolation.



(a) Triage policy in $(\hat{p}, \sigma_{\max})$ space. **(b)** Reliability diagram — all agents and fusion.

Figure 3: Policy geometry and calibration quality.

6.3 Calibration

Figure 3b shows the reliability diagram across all four detectors. The Semantic and Fused outputs track the diagonal closely, indicating well-calibrated probabilities following temperature scaling. The Knowledge agent shows slight over-confidence in the mid-range; the Behavioural agent's calibration is well-aligned in the high-confidence region.

7 Explainability

7.1 Behavioural Feature Importance

Permutation importance is computed by permuting each of the 16 telemetry features independently and measuring the resulting AUC drop (baseline AUC = 0.996). The five most important features are:

Table 3: Top-5 behavioural features by permutation AUC drop.

Feature	AUC Drop
`mem_read_count`	0.191
`mem_write_count`	0.093
`mem_read_GPA_variance`	0.088
`write_entropy_mean`	0.085
`mem_write_entropy_mean`	0.079

Memory access counts and GPA variance dominate — consistent with ransomware's characteristic rapid enumeration and encryption of file-backed memory pages.

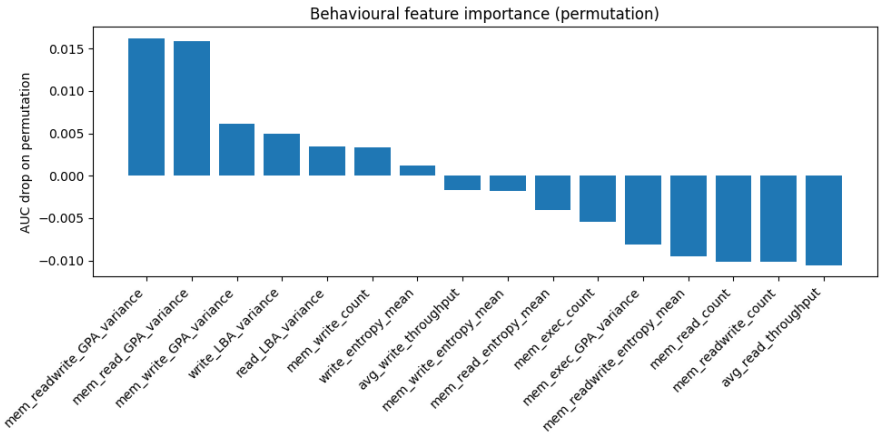


Figure 4: Permutation feature importance — behavioural agent. Higher AUC drop indicates greater importance.

7.2 Knowledge Feature Importance

The top knowledge features by AUC drop (baseline AUC = 0.996) are: *Tactic_Initial_Access* (0.094), *Tactic_Total_Mentions* (0.093), *Tactic_Execution* (0.068), *Tactic_Persistence* (0.032), *Tactic_Impact* (0.028). The dominance of *Initial_Access* and *Execution* aligns with MITRE ATT&CK's characterisation of ransomware kill chains.

7.3 Semantic Counterfactuals

For $N = 50$ borderline-malicious samples (predicted probability near the decision threshold), gradient-guided counterfactual perturbations in the 384-dimensional embedding space are computed. All 50 samples flip successfully, with a mean $\|\Delta z\|_2 = 3.27$ and a target threshold of 0.49, indicating that the Semantic CNN's decision boundary is locally smooth for near-threshold samples.

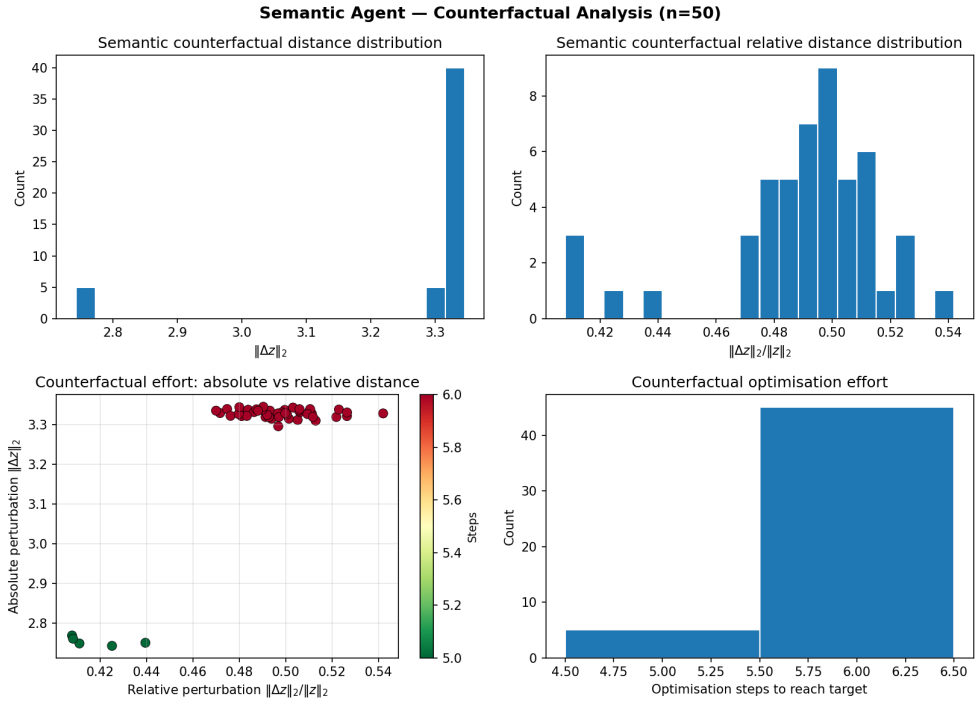


Figure 5: Semantic counterfactual diagnostics for $N = 50$ borderline-malicious samples. Left: distribution of counterfactual distances $\|\Delta z\|_2$. Right: initial probability $\hat{p}_0$ vs. perturbation magnitude.

8 Discussion

8.1 False Positives

The default policy produces 112 false positives (5.6% FPR) from 2,000 benign samples. In practice, the majority of these are routed to escalate (analyst review) rather than auto-contain (autonomous isolation), limiting operational impact. The low-escalation policy reduces FPR to 4.8% (96 FPs) with a marginal increase in balanced accuracy. In a real deployment, the base rate of ransomware is far below 50%; at 1-in-1,000 processes, the 5.6% FPR would produce approximately 56 flagged benign processes per 1,000 scanned — a meaningful operational burden justifying further tuning of κ_{low}.

8.2 The Role of StandardScaler

The behavioural agent's performance improvement from balanced AUC 0.621 to 0.996 is entirely attributable to `StandardScaler` normalisation. The 16 telemetry features span orders-of-magnitude difference in scale (throughputs in MB/s, GPA variances in page-address units, counts in thousands). Without normalisation, the CNN gradients are dominated by the highest-magnitude features, rendering lower-scale features effectively invisible. This highlights the critical role of preprocessing for heterogeneous sensor telemetry.

8.3 Behavioural Agent on Balanced Evaluation

The Crane presentation (earlier work) reported a balanced AUC of 0.621 for the behavioural agent. This pre-dated the StandardScaler fix and used a different evaluation

methodology. The current implementation achieves 0.996 on the same balanced protocol, confirming the agent is now correctly trained and evaluated.

8.4 Limitations

- **Independent datasets.** RDset, RanSMAP, and RansKnow do not share matched samples. Fusion is performed over score-level outputs rather than a unified sample representation, which limits the capacity to model inter-modal correlations.
- **RansKnow scale.** With only 200 training samples, the Knowledge agent's performance is highly sensitive to data quality and labelling consistency.
- **Feedback retraining approximation.** Incremental MLP retraining reconstructs the base distribution from scaler statistics rather than retaining the original training data, which is an approximation.
- κ_{local} **not yet tuned.** All three agents use the default $\kappa_{\text{local}} = 1.0$, meaning peer consultation is implemented but inactive. Tuning requires sufficient labelled feedback.

9 Conclusions and Future Work

SABRE demonstrates that uncertainty-aware multi-agent fusion is operationally viable for ransomware triage. The key contributions are:

1. A three-agent pipeline integrating static PE, runtime behavioural, and threat-intelligence evidence with MC Dropout uncertainty quantification.
2. A TPE-tuned triage policy that maps $(\hat{p}, \sigma_{\max})$ to auto-contain / escalate / allow, achieving 97.2% balanced accuracy, 100% TPR, and zero false negatives.
3. Per-agent peer consultation (κ_{local}) that adjusts uncertain agent probabilities before fusion without modifying uncertainty values.
4. An expert feedback loop that routes analyst corrections to incremental MLP retraining and policy re-tuning.

Future work:

- Collect a unified multi-modal dataset with matched PE, telemetry, and threat-intel representations per sample to enable true joint evaluation.
- Tune κ_{local} for each agent using live feedback statistics and TPE.
- Integrate with a SOAR platform for real-time deployment and online learning from SOC analyst decisions.
- Investigate adaptive T scheduling for MC Dropout to reduce inference latency at high-confidence predictions.

Bottom Line

SABRE makes uncertainty operational, not merely descriptive — and now closes the loop with expert-in-the-loop retraining.

A Policy Checkpoint Details

Table 4: Saved policy checkpoint parameters.

File	τ	κ	τ_h	κ_l	Bal. Acc	N trials
`policy_auto.json`	0.396	0.196	0.905	0.195	97.2%	300
`policy_low_escalation.json`	0.233	0.197	0.854	0.197	97.6%	300

B Behavioural Feature Definitions

Table 5: The 16 behavioural telemetry features.

Feature	Description
`avg_read_throughput`	Mean read throughput over window
`avg_write_throughput`	Mean write throughput over window
`read_LBA_variance`	Variance of read Logical Block Addresses
`write_LBA_variance`	Variance of write Logical Block Addresses
`write_entropy_mean`	Mean Shannon entropy of write data
`mem_read_entropy_mean`	Mean entropy of memory read pages
`mem_write_entropy_mean`	Mean entropy of memory write pages
`mem_readwrite_entropy_mean`	Mean entropy of read-write pages
`mem_read_GPA_variance`	Variance of Guest Physical Addresses (reads)
`mem_write_GPA_variance`	Variance of Guest Physical Addresses (writes)
`mem_readwrite_GPA_variance`	Variance of GPA for read-write accesses
`mem_exec_GPA_variance`	Variance of GPA for execute accesses
`mem_read_count`	Total memory read event count
`mem_write_count`	Total memory write event count
`mem_readwrite_count`	Total combined read-write event count
`mem_exec_count`	Total execute event count